Light

GCSE Computer Science – Networks Basics (Set 1)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a computer network?

- ○ **A.** Two or more computers connected together to share data and resources
- ○ **B.** A single computer connected to a printer
- ○ **C.** A computer with multiple monitors attached
- ○ **D.** Software that allows computers to run faster

Q2 (1 mark)

Which statement best describes a Local Area Network (LAN)?

- ○ **A.** A network that spans multiple countries
- ○ **B.** A network that uses only satellite connections
- ○ **C.** A network that only connects mobile phones
- ○ **D.** A network covering a small geographical area, often owned by a single organisation

Q3 (1 mark)

What is the largest example of a Wide Area Network (WAN)?

- ○ **A.** The internet
- ○ **B.** A school network
- ○ **C.** A home Wi-Fi network
- ○ **D.** A Bluetooth connection between devices

Q4 (1 mark)

Which technology is used to create a Personal Area Network (PAN)?

- ○ **A.** Fibre optic cables
- ○ **B.** Satellite dishes
- ○ **C.** Telephone lines
- ○ **D.** Bluetooth

Q5 (1 mark)

What is one advantage of connecting computers in a network?

- ○ **A.** Resources such as printers and internet connections can be shared
- ○ **B.** Files cannot be shared between computers
- ○ **C.** Each computer must have its own printer
- ○ **D.** Computers become slower when networked

Q6 (1 mark)

What is a network protocol?

- ○ **A.** A set of rules that govern how data is transmitted across a network
- ○ **B.** A type of network cable
- ○ **C.** A device that connects networks together
- ○ **D.** A program for browsing the internet

Q7 (1 mark)

What is HTTP used for?

- ○ **A.** Sending and receiving emails
- ○ **B.** Connecting Bluetooth devices

- ○ **C.** Converting digital signals to analogue
- ○ **D.** Transferring web pages between a web server and a browser

Q8 (1 mark)

How does HTTPS differ from HTTP?

- ○ **A.** HTTPS is faster than HTTP
- ○ **B.** HTTPS provides an encrypted version of HTTP for more secure web transactions
- ○ **C.** HTTPS only works on mobile devices
- ○ **D.** HTTPS is used for emails while HTTP is used for websites

Q9 (1 mark)

What is the main purpose of a firewall?

- ○ **A.** To speed up internet connections
- ○ **B.** To monitor network traffic and block or allow it based on security rules
- ○ **C.** To store backup copies of files
- ○ **D.** To convert wireless signals to wired signals

Q10 (1 mark)

Which is an advantage of wireless networks compared to wired networks?

- ○ **A.** Wireless networks are always faster
- ○ **B.** Wireless networks are more secure
- ○ **C.** Wireless networks never experience interference
- ○ **D.** Devices can connect without needing physical cables

Q11 (1 mark)

At which layer of the TCP/IP model do web browsers and email programs operate?

- ○ **A.** Application layer
- ○ **B.** Transport layer
- ○ **C.** Internet layer
- ○ **D.** Link layer

Q12 (1 mark)

What does MAC address filtering control?

- ○ **A.** The speed of data transfer on the network
- ○ **B.** Which websites users can visit
- ○ **C.** The encryption level of transmitted data

○ **D.** Which devices can access or be blocked from accessing a network

Q13 (2 marks)

A company has offices in London and Manchester. Each office has its own internal network. The company wants to connect these two networks so employees can share files between offices. What type of network connection would link the London and Manchester offices together?

○ **A.** A Personal Area Network (PAN)

○ **B.** A Local Area Network (LAN)

○ **C.** A Bluetooth connection

○ **D.** A Wide Area Network (WAN)

Q14 (2 marks)

A business is choosing between wired and wireless networking for a new office. Which of the following is a disadvantage of choosing wireless networking?

○ **A.** Wireless networks are more susceptible to interference and security risks

○ **B.** Wireless networks cannot connect to the internet

○ **C.** Wireless networks require cables to be installed throughout the building

○ **D.** Wireless networks do not allow mobile devices to connect

Q15 (2 marks)

A school is installing a new network. The network administrator must choose between fibre optic cables and copper cables for the main network connections. In which situation would fibre optic cable be the better choice?

- ○ **A.** When the budget is very limited and low cost is the priority
- ○ **B.** When connecting devices that are very close together in the same room
- ○ **C.** When very high-speed data transfer over longer distances is required
- ○ **D.** When the network only needs to support a few users

Q16 (2 marks)

When data is sent across the internet, both TCP and IP protocols are used. What are their respective roles?

- ○ **A.** TCP displays web pages and IP sends emails
- ○ **B.** TCP manages the reliable delivery of data, while IP handles addressing and routing data packets
- ○ **C.** TCP creates encryption and IP provides passwords
- ○ **D.** TCP connects to Wi-Fi and IP connects to wired networks

Q17 (2 marks)

A user sends an email from their computer and later reads a reply on their smartphone. Which protocols are involved in sending the email and retrieving the reply?

- ○ **A.** HTTP for sending and HTTPS for retrieving
- ○ **B.** SMTP for sending and IMAP for retrieving
- ○ **C.** TCP for sending and IP for retrieving
- ○ **D.** IP for sending and TCP for retrieving

Q18 (2 marks)

A company uses both encryption and authentication to protect its network. How do these two security methods work together?

- ○ **A.** Encryption speeds up the network while authentication slows it down
- ○ **B.** Authentication verifies who is accessing the network, while encryption protects data from being read if intercepted
- ○ **C.** Encryption prevents all viruses while authentication blocks spam emails
- ○ **D.** Authentication creates backups while encryption deletes old files

Q19 (2 marks)

In the TCP/IP model, what is the main function of the internet layer?

- ○ **A.** To run web browsers and email applications
- ○ **B.** To establish communication between hosts and agree packet sizes
- ○ **C.** To address and package data for transmission, and route packets across the network

○ **D.** To manage the physical network hardware such as network interface cards

Q20 (2 marks)

A school is considering connecting all its computers to a network. Which statement correctly identifies both an advantage and a disadvantage of doing this?

○ **A.** Advantage: Each computer needs its own printer. Disadvantage: Files can be easily shared.

○ **B.** Advantage: Software can be installed centrally. Disadvantage: If the server fails, users may lose access to shared resources.

○ **C.** Advantage: The network makes computers slower. Disadvantage: Security is improved.

○ **D.** Advantage: Each computer works independently. Disadvantage: Resources cannot be shared.

Q21 (2 marks)

At which layer of the TCP/IP model would you find the network interface card (NIC) and device drivers?

○ **A.** Application layer

○ **B.** Transport layer

○ **C.** Internet layer

○ **D.** Link layer

Q22 (2 marks)

A network administrator uses both a firewall and MAC address filtering to protect the network. How do these methods provide different types of protection?

- ○ **A.** They provide exactly the same protection, so only one is needed
- ○ **B.** The firewall controls which devices can connect based on hardware address, while MAC filtering monitors data traffic
- ○ **C.** The firewall monitors and filters network traffic based on rules, while MAC filtering controls which specific devices can connect based on their hardware address
- ○ **D.** The firewall encrypts all data, while MAC filtering creates passwords

Q23 (2 marks)

A small office is setting up a network to connect five computers in the same room. Why might copper cables be a more appropriate choice than fibre optic cables for this situation?

- ○ **A.** Copper cables are faster than fibre optic cables
- ○ **B.** Copper cables can transmit data over longer distances than fibre
- ○ **C.** Copper cables are less expensive and provide sufficient speed for short distances with few users
- ○ **D.** Copper cables are immune to electrical interference

Q24 (3 marks)

A hospital is implementing network security to protect patient records. The security team decides to use authentication, encryption, a firewall,

and MAC address filtering together. Explain why using multiple security methods provides better protection than using just one.

- ○ **A.** Using multiple methods is unnecessary because each method provides complete protection on its own
- ○ **B.** Multiple methods slow down the network so much that hackers give up trying to access it
- ○ **C.** Each method protects against different threats, so together they provide layered defence—if one method is bypassed, others still provide protection
- ○ **D.** Multiple methods are only needed for home networks, not hospital networks

Q25 (3 marks)

When a user sends an email, the message passes through all four layers of the TCP/IP model. Starting from the application layer, describe the correct order of layers and what happens at each layer.

- ○ **A.** Link layer → Internet layer → Transport layer → Application layer: data moves from physical hardware up to the email program
- ○ **B.** Application layer → Transport layer → Internet layer → Link layer: email is prepared, broken into packets, addressed, then transmitted physically
- ○ **C.** Transport layer → Application layer → Link layer → Internet layer: packets are created first, then the email is written
- ○ **D.** Internet layer → Link layer → Application layer → Transport layer: routing happens first, then the email is created

Q26 (3 marks)

A student uses Bluetooth to connect wireless headphones to their laptop at home. Their home has a Wi-Fi network connecting all devices to the

internet. They also access their school's learning platform, which is hosted on a server in another city. Identify the three different types of networks involved in this scenario.

- ○ **A.** Three PANs: one for headphones, one for Wi-Fi, one for the school server
- ○ **B.** Three LANs: one for headphones, one for home network, one for school access
- ○ **C.** PAN (Bluetooth headphone connection), LAN (home Wi-Fi network), WAN (connection to school server in another city)
- ○ **D.** Three WANs: all connections involve the internet

Q27 (3 marks)

A library is installing a new network and must decide between wired and wireless networking. The library has fixed desktop computers at research stations, mobile tablet devices for staff, and areas where visitors use their own laptops. Evaluate which type of networking would be most appropriate.

- ○ **A.** Wired only—all devices should use cables for maximum security
- ○ **B.** Wireless only—cables are outdated and should never be used
- ○ **C.** A combination: wired connections for fixed desktop computers (reliable and secure) and wireless for mobile devices and visitor laptops (flexibility)
- ○ **D.** Neither—libraries should not have computer networks

Q28 (3 marks)

A user visits an online shopping website. Initially, the browser shows 'http://' in the address bar. When the user goes to the payment page to

enter credit card details, the address changes to 'https://'. Explain why this change is important and what protection HTTPS provides.

- ○ **A.** HTTPS makes the website load faster, which is useful for payment pages
- ○ **B.** HTTPS is required because payment pages have more images to download
- ○ **C.** HTTPS encrypts the data transmitted between the browser and server, protecting sensitive payment information from being intercepted and read by attackers
- ○ **D.** HTTPS changes the website's appearance to look more trustworthy

Q29 (3 marks)

A university is upgrading its network. The main connection between campus buildings is currently copper cable and experiences slow speeds and occasional interference from nearby electrical equipment. The IT department is considering upgrading to fibre optic cable. Evaluate the advantages of this upgrade.

- ○ **A.** Fibre optic is cheaper than copper, which is the main advantage
- ○ **B.** Fibre optic provides higher speeds over longer distances and is immune to electrical interference, solving both problems the university is experiencing
- ○ **C.** Fibre optic only works for wireless networks, which is what the university needs
- ○ **D.** Fibre optic requires less maintenance because it never breaks

Q30 (3 marks)

A school is implementing a new network security policy. Currently, students can connect any device to the school Wi-Fi. The school wants to improve security. The IT manager proposes using authentication (username and password for each student) combined with MAC address filtering. Evaluate how this combined approach improves security.

- ○ **A.** This approach provides no improvement because MAC addresses cannot be checked
- ○ **B.** Authentication alone would be sufficient; MAC filtering adds no benefit
- ○ **C.** MAC filtering provides two security layers: MAC filtering ensures only school-registered devices can connect, while authentication verifies the student's identity—an attacker would need both an approved device AND valid credentials
- ○ **D.** This approach would prevent students from using the internet entirely

Total: **55** marks • Time: **30 min**

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